



Molecular dynamics interaction study on natural DPP4-inhibitors

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Abstract

Diabetes mellitus is one of the human diseases with the highest morbidity and mortality rate in the world, being of the utmost importance to find alternative treatments to manage the complications of the disorder. DPP4 inhibitors prevent the degradation of GLP-1, thereby enhancing insulin secretion and compensating for insulin deficiency. In this article, through molecular engineering, we are able to analyse the binding affinity of potential natural DPP4 inhibitors by dynamic interactions and their ADMET toxicity profiles of the ligands as potential DPP4 inhibitors. In our study, twenty two compounds were selected for blind docking simulations, the five best-ranked binding ligands proceeded with subsequent molecular dynamics simulations.

Introduction

Diabetes mellitus is a chronic metabolic disorder characterized by persistent hyperglycemia resulting from insulin resistance or defects in insulin secretion and/or action, and it is associated with high morbidity, mortality, and a substantial global economic burden^{1,2}. Dipeptidyl Peptidase-4 (DPP-4) inhibitors are known for their ability to³⁻⁵ prolong the activity of glucagon-like peptide-1 (GLP-1), thereby enhancing insulin secretion and reducing glucagon release³⁻⁵. In general, GLP-1RAs are preferred over DPP-4 inhibitors because they provide greater reductions in HbA1c and induce clinically meaningful weight loss, as demonstrated in clinical trials. In contrast, DPP-4 inhibitors produce only modest increases in endogenous GLP-1 levels, resulting in smaller glycemic improvements and minimal effects on body weight⁶. In this context, DPP-4⁷ have gained increasing attention due to their role in glucose homeostasis and the development of novel pharmacological strategies^{1,8}. In this study, we aim to identify natural compounds that can act as DPP-4 inhibitors using molecular engineering techniques, employing molecular modeling and docking approaches to screen for natural compounds that may interact with DPP-4, predicting their binding affinities and potential effects. This approach aims to discover low-cost, natural alternatives for diabetes treatment, offering new therapeutic options for managing the disease and its complications.

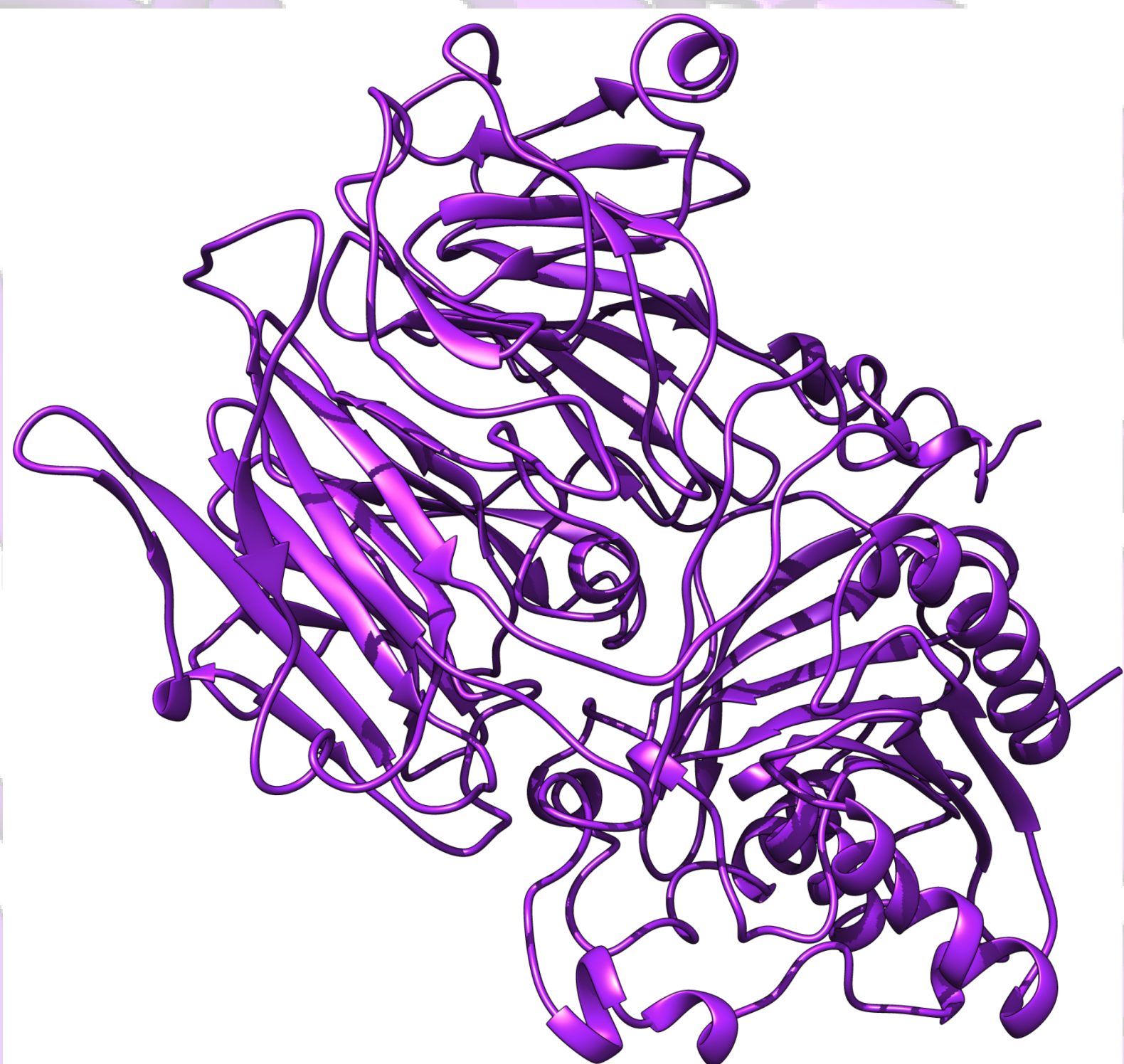


Figure 2: Final DPP4 structure.

Methodology

The present study focuses on several platforms and software tools, including the Protein Data Bank (PDB), Chimera, GROMACS, Gaussian, Web of Science, LaTeX, JabRef. The DPP4 and GLP-1 protein structures had already been analyzed using ChimeraX and GROMACS software. A search for known natural ligands resulted in a total of 52 candidate compounds. SMILES notation and three-dimensional structures of the ligands were retrieved and verified through PubChem, Lotus, and Coconut databases. In the research, twenty two natural potential DPP4 inhibitors were selected. The ligands were previously prepared through structural cleaning and geometric optimization using the Gaussian 16 software. The optimized ligands were converted to the PDBQT format for molecular docking. During this step, polar hydrogens were added and nonpolar hydrogens were merged according to AutoDock Vina requirements. Docking simulations were then performed using AutoDock Vina, generating 30 binding poses per ligand with a maximum of 1000 steps per run. The 5 best-ranked binding ligands were selected based on binding affinity scores from the DPP4 ligands (isoquercitrin (L1), rutin (L2), a-murolene (L3), marmesinin (L4) and a flavone derivative (L5)), with values between -10.031 to -8.209 kcal/mol, which shows they had a good binding potential.

These selected complexes were further processed in Chimera, where hydrogens were added to the ligands to ensure proper protonation. Ligand topologies and parameters were then generated using the LigParGen server, which provided OPLS-compatible force field parameters, including ligand topology files (.itp), and structure files (.gro), for subsequent molecular dynamics simulations. Hirshfeld atomic charges were calculated for the ligands using Gaussian 16. The resulting charges were adjusted to ensure overall charge neutrality of each ligand and were subsequently incorporated into the ligand topology files for molecular dynamics simulations. After that, molecular dynamics simulations were carried out using GROMACS. The systems were subjected to energy minimization, followed by equilibration under NVT and NPT ensembles, and then to steered molecular dynamics (SMD) simulations. RMSD and RMSF analyses were performed on the GROMACS trajectories to evaluate the global stability and residue-wise flexibility of the protein-ligand complexes. Binding free energy calculations were then conducted using the Molecular Mechanics Poisson-Boltzmann Surface Area (MM/PBSA) method based on simulation snapshots to obtain refined binding affinity estimates beyond static docking performed in gmx MMPBSA.

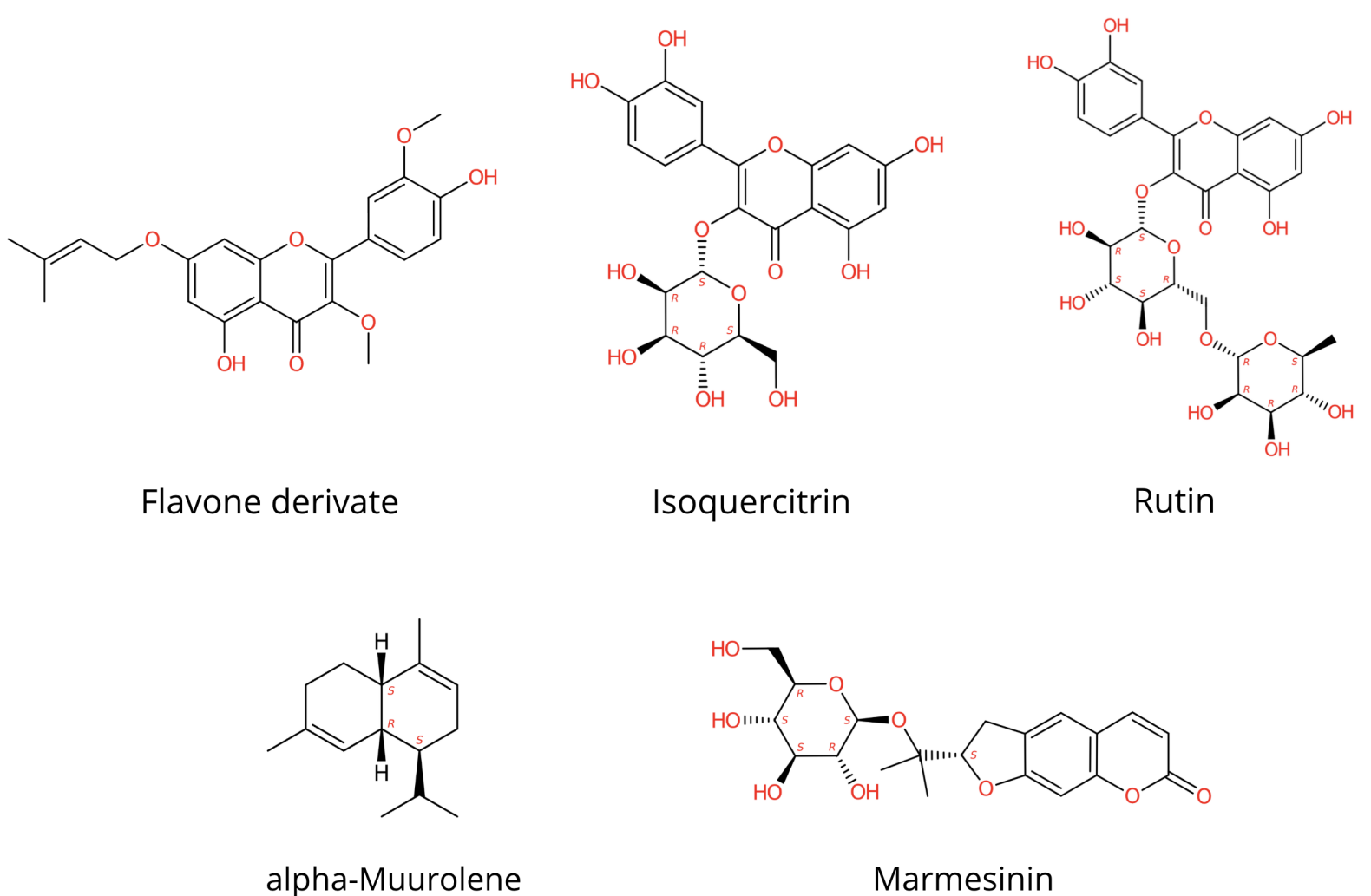


Figure 4: Natural ligands selected for the dynamics.

Results and Discussion

Figure 2 shows the final stable state of the DPP4 and Figure 8 shows the RMSD and RMSF of this equilibrium showing an excellent stable conformation. Figure 10 shows that the stable state of the protein (RMSF).

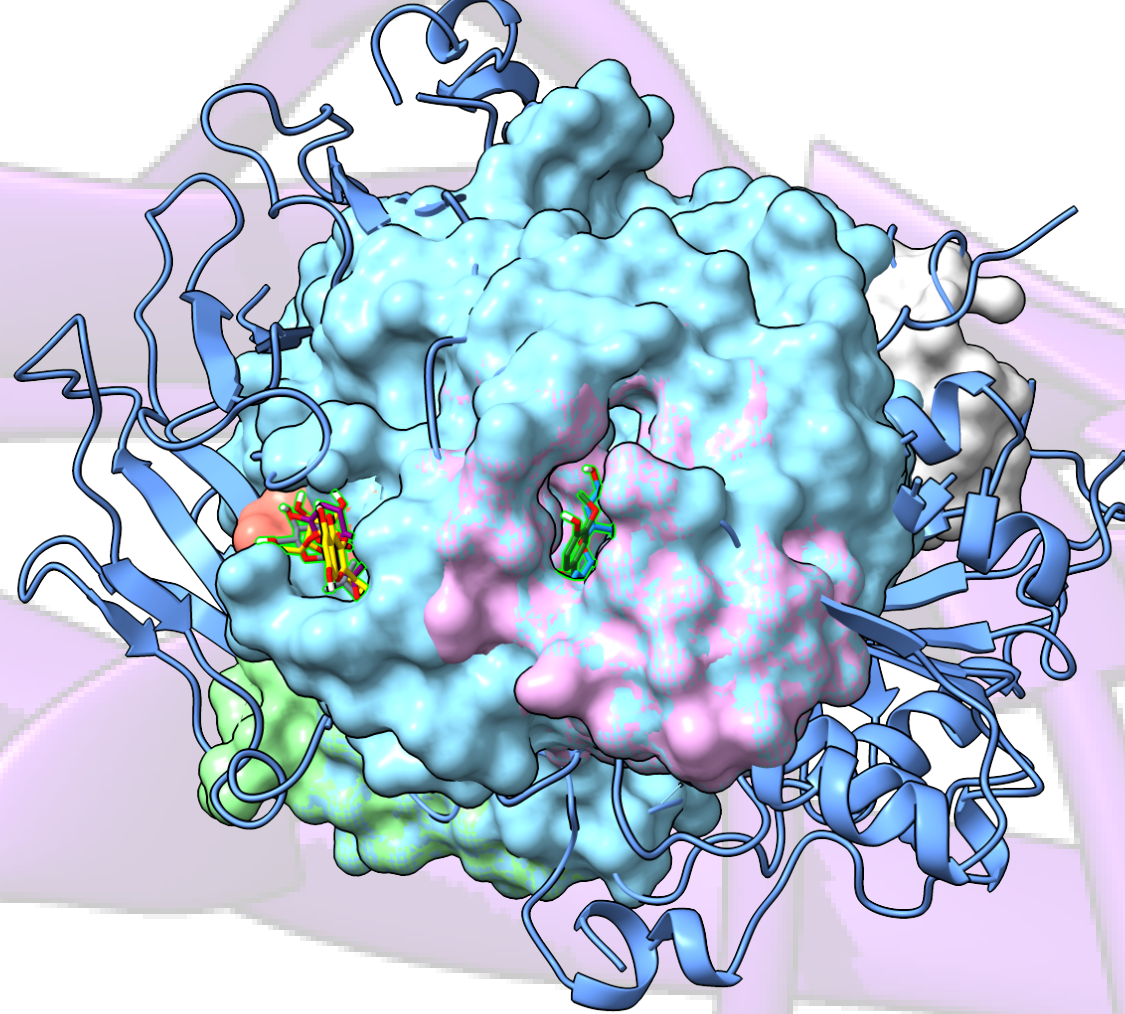


Figure 6: DPP4 500ns equilibrium NPT.

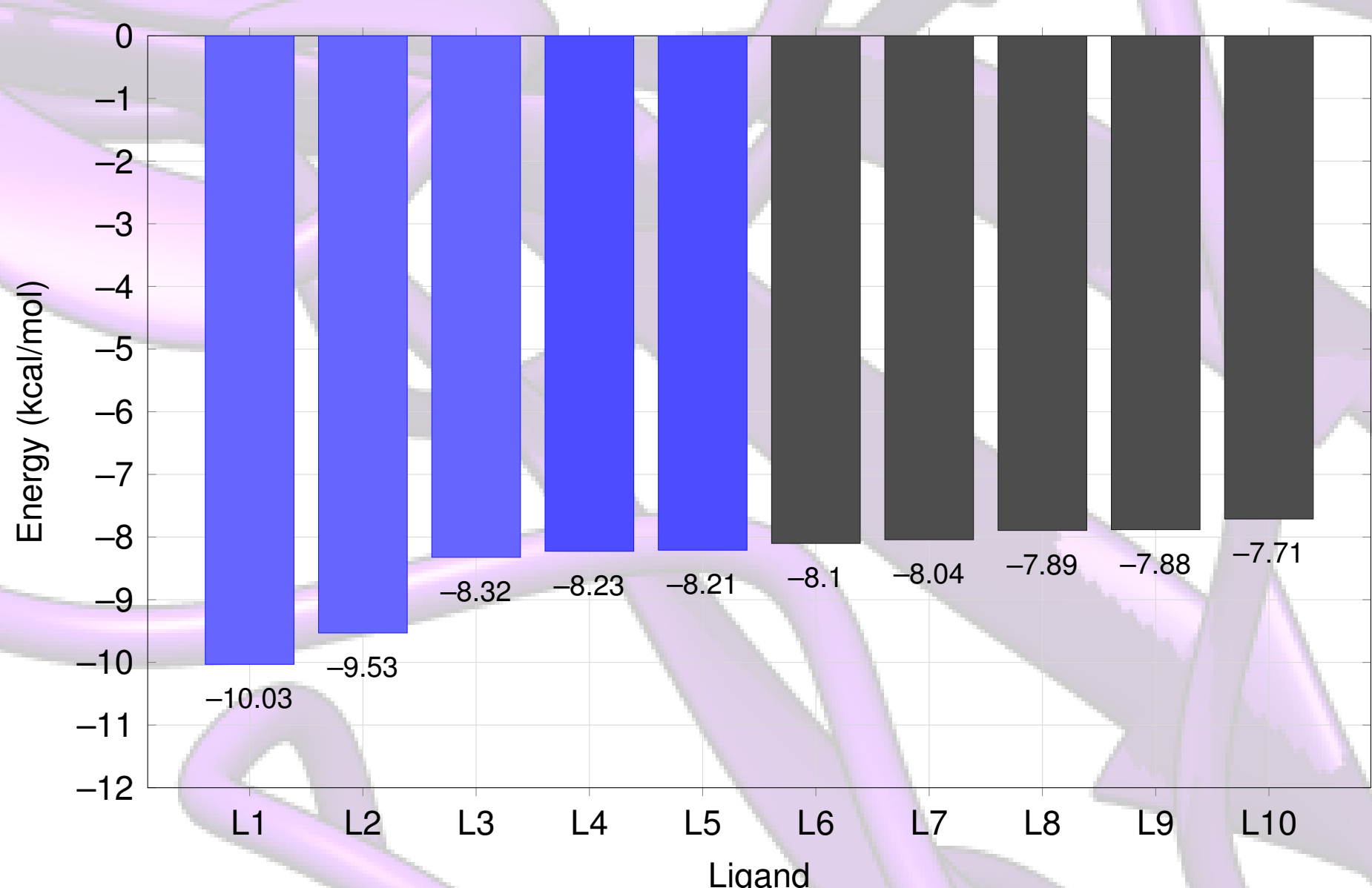


Figure 8: The 10 best results after blind docking.

During the RMSD and RMSF analysis, shown in Figure 12 it was observed that L1 (isoquercitrin) destabilized the protein the most between 40 ns to 60ns. The other 4 ligands kept the protein in a fairly stable state.

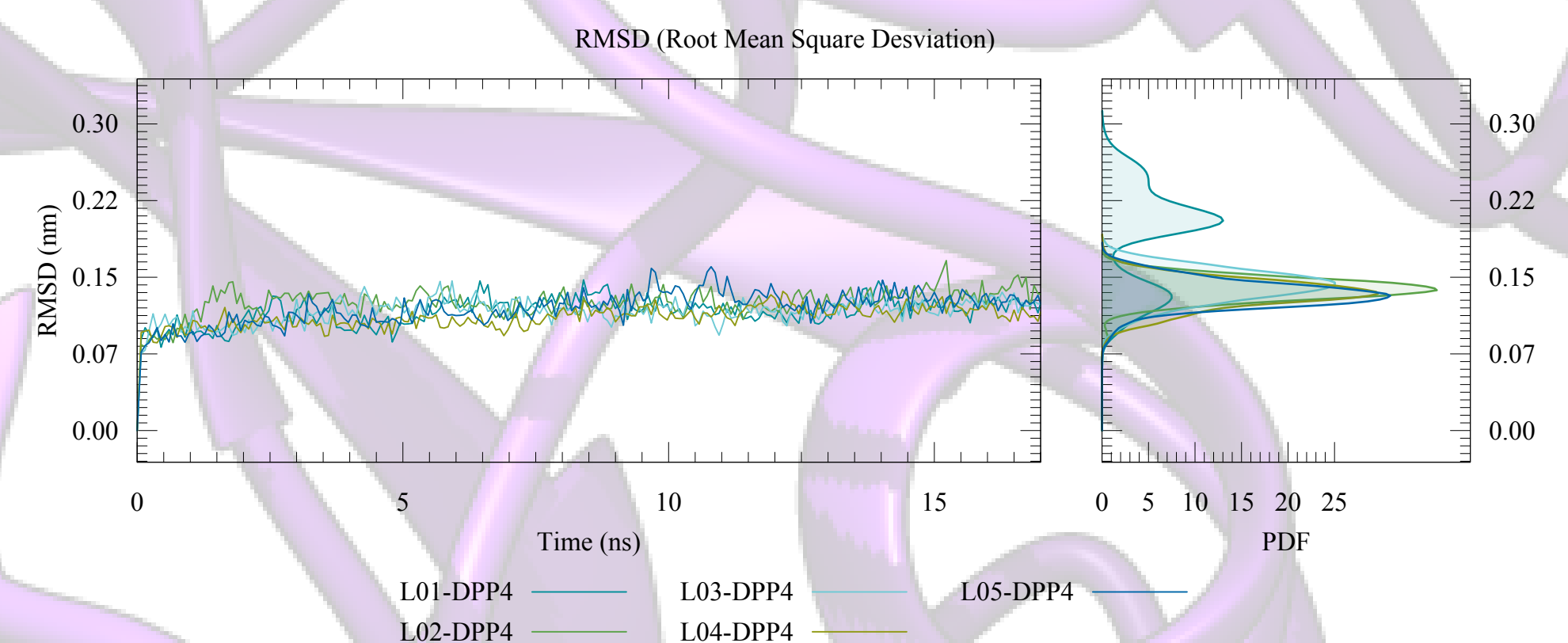


Figure 10: DPP4-ligands RMSD (top) and RMSF (bottom) after 100 ns.

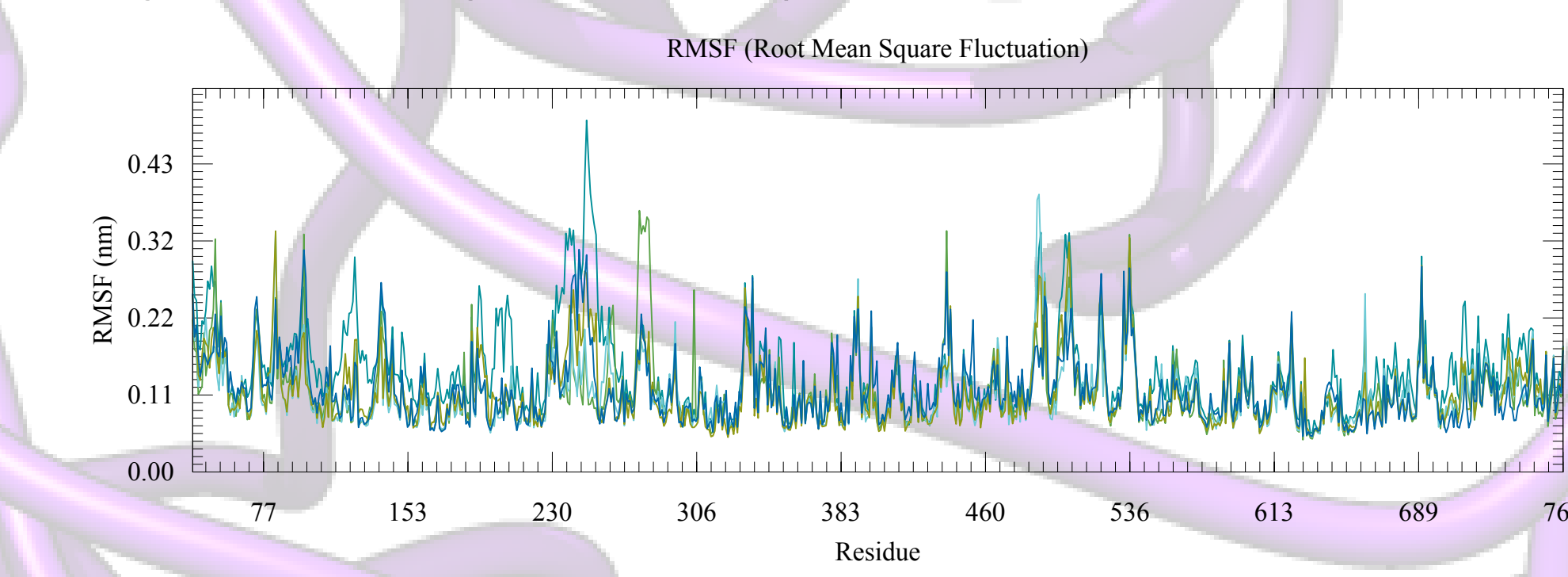


Figure 12: DPP4-ligands RMSD (top) and RMSF (bottom) after 100 ns.

The reactivity of each ligand was analyzed and as shown in Figure 14. L5 (a flavone derivative) displayed the most negative energy out of all of them, meaning that it is the most reactive. L4 (marmesinin) had the second best reactivity, with a very small difference from L5.

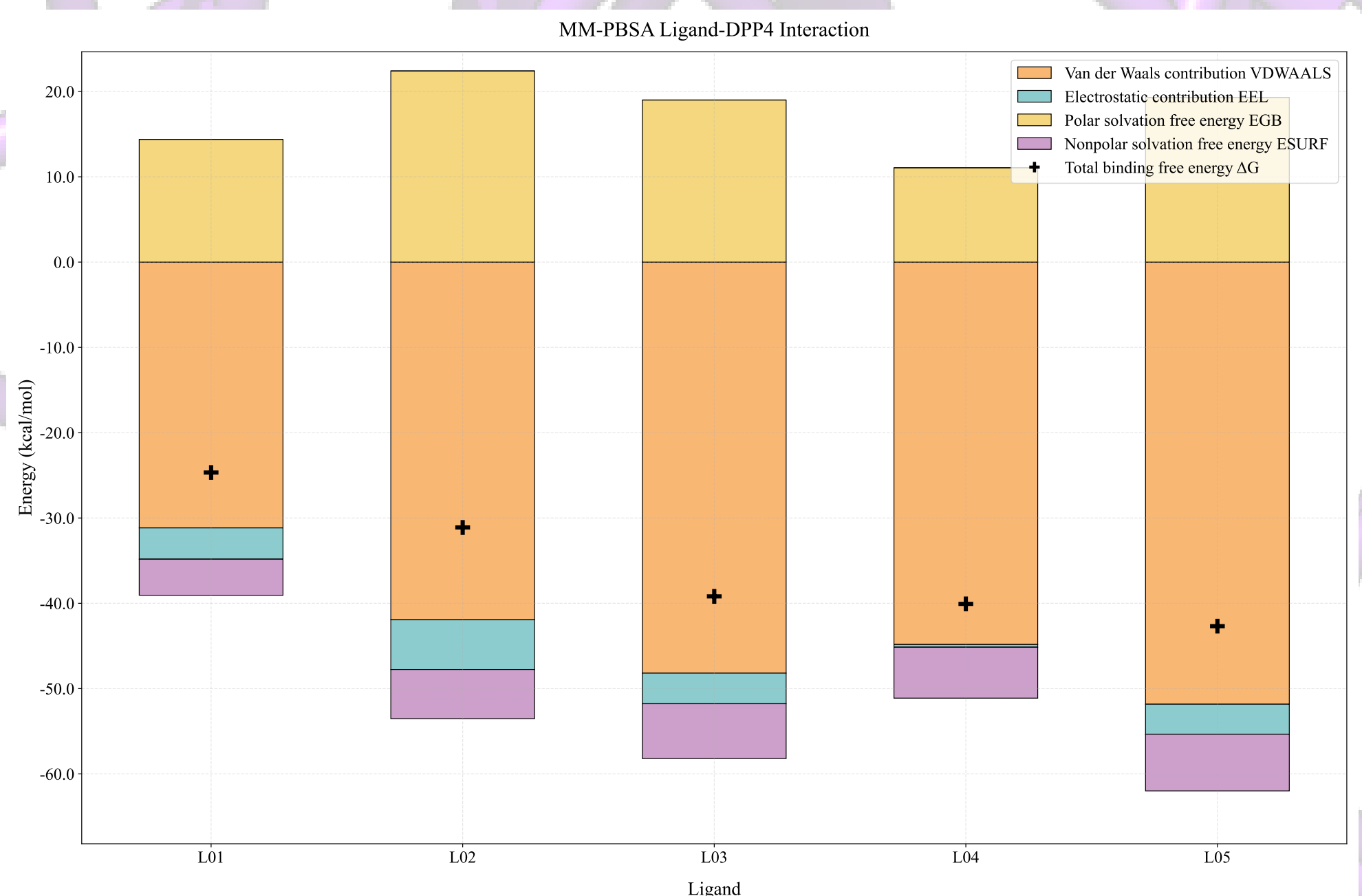


Figure 14: DPP4-ligands MMPBSA energy decomposition.

Compound	ΔG _{bind}
Isoquercitrin (L1)	-24.67
Rutin (L2)	-31.11
A-murolene (L3)	-39.19
Marmesinin (L4)	-40.07
Flavone derivative (L5)	-42.68

The toxicity profiles of the top five ligands were evaluated using *in silico* ADMET 3.0 analysis, revealing heterogeneous patterns. All compounds displayed high skin sensitization potential. Rutin (L2) emerged as the most toxic, showing strong hepatotoxic, mutagenic, and cytotoxic tendencies. Isoquercitrin also presented significant genotoxicity, while 5,4'-Dihydroxy-3,3'-dimethoxy-7-prenyloxyflavone

(L5) exhibited moderate hepatotoxic and carcinogenic risks. α-Murolene (L3) showed a narrower toxicity profile, primarily affecting skin and eyes, whereas Marmesinin (L4) demonstrated considerable hepatotoxic and genotoxic potential. These findings suggest that the α-Murolene is the safest one based on the AD-MET analysis.

Conclusions

This study evaluated the interaction of natural ligands with GLP-1, a key target in diabetes treatment. Following blind molecular docking, 5 potential candidates were chosen based on their binding energies and after molecular dynamics simulations it was revealed that 5,4'-Dihydroxy-3,3'-dimethoxy-7-prenyloxyflavone, a flavone derivative had the best reactivity with DPP4 and exhibited moderate hepatotoxic and carcinogenic risks, being the most promising candidate for further optimization. Marmesinin showed the second best reactivity, despite its high AD-MET toxicity. Alfa-murolene showed the third best reactivity, with the least toxicity profile, exhibiting a good potential for further investigation. However, isoquercitrin and rutin, despite their good binding affinities, demonstrated significant toxicity risks, limiting their therapeutic potential. In conclusion, this study shows possible therapeutic candidates for DPP4 inhibition in diabetes treatment, integrating dynamic studies and toxicity profiling in selecting potential DPP4 inhibitors for diabetes treatment. Further experimental validation will be essential to confirm these findings.

Acknowledgement

This study was conducted under the surveillance of Professor Badhin, in the computational laboratory of the Catholic University of Santa Maria. We would like to thank Professor Badhin for the hospitality and kindness, allowing the realization of the internship and to learn about molecular engineering, its importance, and the opportunity to grow in an enriching research environment. We are also grateful for the computational infrastructure that facilitated our analysis. We thank our tutors and colleagues for their patience invaluable guidance, feedback, and collaboration.

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